

**Problem 1** (§2.1, #2,4,8). Decide whether or not the following are statements. In the case of a statement, say if it is true or false, if possible.

- a) Every even integer is a real number.
- b) Sets  $\mathbb{Z}$  and  $\mathbb{N}$ .
- c)  $\mathbb{N} \notin \mathcal{P}(\mathbb{N})$ .

**Problem 2** (§2.2, #2,6,8 - adapted). Express each statement or open sentence in a symbolic form such as  $P \wedge Q$ ,  $P \vee Q$ ,  $P \vee (\sim Q)$ , or  $(\sim P)$ , etc. Be sure to also state exactly what statements  $P$  and  $Q$  stands for. *Remark: You should not use any quantifiers.*

- a) The matrix  $A$  is not invertible.
- b) There are quizzes scheduled for Tuesday and Thursday.
- c) At least one of the numbers  $x$  and  $y$  equals 0.

**Problem 3** (§2.3, #2,8). Without changing their meanings, convert each of the following sentences into a sentence having the form “If  $P$ , then  $Q$ ”.

- a) For a function to be continuous, it is sufficient that it is differentiable.
- b) A geometric series with ratio  $r$  converges if  $|r| < 1$ .

**Problem 4** (§2.6, #8). Use a truth table to show that the following statements are logically equivalent.

$$(\sim P \iff Q) = ((P \implies \sim Q) \wedge (\sim Q \implies P))$$

**Problem 5** (§2.7, #6,8). Write the following as English sentences. Say whether they are true or false.

- a)  $\exists n \in \mathbb{N}, \forall X \in \mathcal{P}(\mathbb{N}), |X| < n$ .
- b)  $\forall n \in \mathbb{Z}, \exists X \subseteq \mathbb{N}, |X| = n$ .

**Problem 6** (§2.9, #6). Translate the following sentence into symbolic logic.

“For every positive number  $\varepsilon$  there is a positive number  $M$  for which  $|f(x) - b| < \varepsilon$ , whenever  $x > M$ .”

**Problem 7** (§2.10, #6 - adapted). Write in symbolic language and negate the following sentence.

“There exists a real number  $a$  for which  $a + x = x$  for every real number  $x$ .”